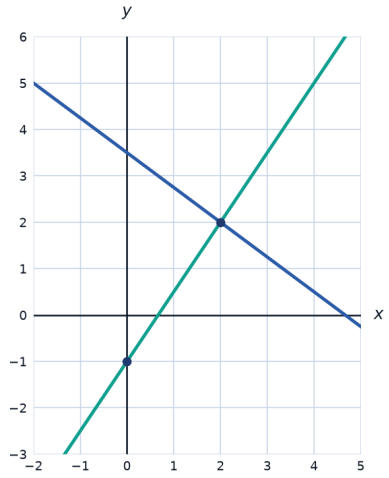
# Solving contextual problems with integers

## Core Concepts: Solving contextual problems with integers

**1** For the model $f(x) = 4x + 5$ in solving contextual problems with integers, evaluate $f(2)$.




**a.** 13

**b.** 17

**c.** 8

**d.** 17

**2** Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

**a.** $x = 1$ and $x = 5$

**b.** $x = 1$ and $x = 4$

**c.** $x = -1$ and $x = -4$

**d.** $x = 5$ and $x = 4$

**3** A data set recorded for solving contextual problems with integers consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

**a.** 15.5

**b.** 16.5

**c.** 17.5

**d.** 19.5

## Procedural Fluency & Algebraic Manipulation

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**4** For the model $f(x) = 4x + 5$ in solving contextual problems with integers, evaluate $f(3)$.

**a.** 17

**b.** 21

**c.** 12

**d.** 21

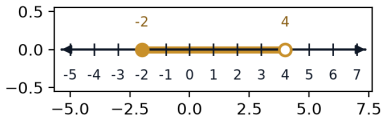
**5** Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

**a.** $x = 2$ and $x = 6$

**b.** $x = 2$ and $x = 5$

**c.** $x = -2$ and $x = -5$

**d.** $x = 7$ and $x = 10$




**6** A data set recorded for solving contextual problems with integers consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

**a.** 15.5

**b.** 16.5

**c.** 17.5

**d.** 19.5

## Graphical & Geometric Interpretation

**7** For the model $f(x) = 4x + 5$ in solving contextual problems with integers, evaluate $f(4)$.

**a.** 21

**b.** 25

**c.** 16

**d.** 25

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**8** Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

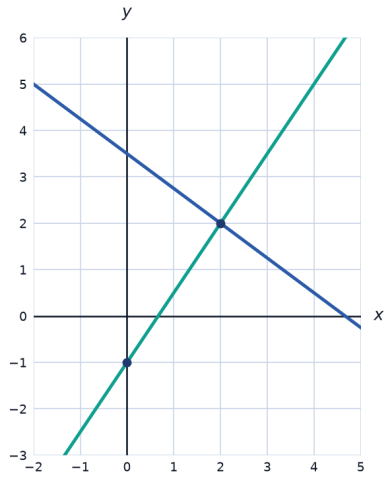
**a.** $x = 3$ and $x = 7$

**b.** $x = 3$ and $x = 6$

**c.** $x = -3$ and $x = -6$

**d.** $x = 9$ and $x = 18$

**9** A data set recorded for solving contextual problems with integers consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?




**a.** 15.5

**b.** 16.5

**c.** 17.5

**d.** 19.5

## Exam-Style Problem Solving & Modeling

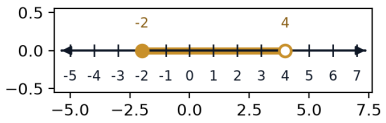
**10** For the model $f(x) = 4x + 5$ in solving contextual problems with integers, evaluate $f(5)$.

**a.** 25

**b.** 29

**c.** 20

**d.** 29




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**11** Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

**a.** $x = 4$ and $x = 8$

**b.** $x = 4$ and $x = 7$

**c.** $x = -4$ and $x = -7$

**d.** $x = 11$ and $x = 28$

**12** A data set recorded for solving contextual problems with integers consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

**a.** 15.5

**b.** 16.5

**c.** 17.5

**d.** 19.5

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